

## PRODUCT LITERATURE

### ANTIPAIN OINTMENT

#### Each 100 g contains

|                   |      |
|-------------------|------|
| Methyl Salicylate | 25 g |
| Camphor           | 2 g  |
| Menthol           | 2 g  |

#### Product Description

Soft white transparent ointment with pleasant odour.

#### Pharmacodynamics

**Camphor:** Camphor is a stimulant, rubefacient, antipruritic and feebly antiseptic. When in combination with other ingredient like menthol, methyl salicylate, it becomes ideal for neuralgia and other painful areas.

**Menthol:** Menthol dilates the blood vessels causing a sensation of coldness, followed by an analgesic effect. Menthol also acts as a penetration enhancer, increasing the penetration of drugs when applied on the skin, to give a faster onset of action.

**Methyl Salicylate:** Methyl Salicylate is a salicylic acid derivative. Salicylates inhibit cyclooxygenase, thereby reducing the formation of prostaglandins, and cause platelet dysfunction. Methyl Salicylate is used topically as a counter – irritant. Upon applications, it is absorbed through the skin and is applied for the relief of pain in rheumatic conditions and painful muscle or joints.

#### Pharmacokinetics

##### Camphor

- Camphor is readily absorbed from all administration sites. Camphor is well absorbed after dermal exposure.
- It is hydroxylated in the liver to yield hydroxycamphor metabolites which are then conjugated with glucuronic acid and excreted in the urine.
- Camphor crosses the placenta.

##### Menthol

- After absorption, menthol is excreted in the urine and bile as a glucuronide.

##### Methyl Salicylate

- May be absorbed through intact skin.
- The absorption of topical salicylates is proportional to the surface area involved, duration of exposure, concentration and skin integrity.
- Absorption characteristics of salicylates vary with the dose, formulation and route of administration.
- Percutaneous absorption is enhanced by exercise, heat, occlusion or disruption of the integrity of the skin or application to large areas of skin.
- Both the rate and extent of absorption increases after repeated application; increasing the bioavailability.
- Methyl salicylate is extensively metabolized to salicylic acid in the dermal and subcutaneous tissues after topical application.
- At therapeutic levels, the half-life of salicylates is 2-4 hours. As salicylate level increase into the toxic range, the half-life can be greater than 18 hours.

#### Indication

For the relief of muscular and rheumatic pains.

#### Recommended Dose

To be massaged well to the affected area, 3 – 4 times daily.

#### Route of Administration

Topical

#### Contraindications

- Do not use in patients with aspirin or salicylate idiosyncrasy
- Hypersensitivity to any of the ingredients in the product
- Product can cause convulsions. Contraindicated in infants below 2 years of age. Caution must be exercised when older children are treated.

#### Warnings and Precautions

Warning and Precautions for camphor:

- This product is contraindicated in children below 2 years of age. Caution must be exercised when older children are treated. Avoid direct application into nostrils.
- It is dangerous to place any camphor containing product into nostril of children. A small amount applied this way may cause immediate collapse.

Warning for Methyl Salicylate

- Topical analgesic preparations containing methyl salicylate should be used with caution in patients at increased risk of developing salicylate adverse effects
- Children suffering from flu, chickenpox, or fever should avoid using this product because salicylates may induce Reyes Syndrome
- This product contains methyl salicylate and when applied or rubbed on to the skin, can be absorbed through the skin into the blood. For patients taking warfarin, excessive application on to the skin for muscle or joint pains may increase the chances of bleeding.

Precautions for Methyl Salicylate:

- Initially apply a small amount to a limited skin area and observe whether there is discomfort or itch before applying to a larger skin area. Discontinue use if irritation develops.
- For external use only; not to be taken.
- Avoid contact with the eyes and mucous membranes.
- Do not apply to open wounds or damaged skin.
- Avoid excessive use during pregnancy and lactation
- Do not apply over large areas of the body or bandage or use with heating pads.

#### Interactions with Other Medicaments

This product contains methyl salicylate and when applied or rubbed on to the skin, can be absorbed through the skin into the blood. For patients taking warfarin, excessive application on to the skin for muscle or joints pains may increase the chances of bleeding.

#### Pregnancy and Lactation

Safety for use in pregnancy has not been established; therefore the potential benefit of the product should be weighed against the possible risks to the mother and child.

**Side Effects**

Ingestion may cause gastrointestinal disturbances such as nausea, dyspepsia and vomiting. Other irritation of the gastric mucosa with ulceration, erosion, haematemesis, melaena and slight blood loss may occur.

This product can cause convulsions.

Headache, dizziness, nausea and vomiting, skin irritation, contact dermatitis, rash, itching, redness or swelling, burning or stinging sensation may occur.

May cause hypersensitivity/allergic reactions in some individuals with sensitive skin.

**Symptoms and Treatment of Overdose**

Symptoms of poisoning includes nausea, vomiting, acidosis, pulmonary edema, pneumonia, convulsions and death.

***Symptoms of methyl salicylate overdose***

Salicylate intoxication can occur after ingestion of topical application of methyl salicylate. Mild chronic salicylate intoxication, or salicylism, usually occurs only after repeated use of large doses. Salicylism can also occur following excessive topical application of salicylates. Symptoms include dizziness, tinnitus, deafness, sweating, nausea and vomiting, headache, and confusion and may be controlled by reducing the dosage.

Symptoms of more severe intoxication or of acute poisoning following overdose include hyperventilation, fever, restlessness, ketosis, and respiratory alkalosis and metabolic acidosis. Depression of the CNS may lead to coma; cardiovascular collapse and respiratory failure may also occur.

***Symptoms of camphor/menthol overdose***

Ingestion of significant quantities is reported to cause symptom: severe abdominal pain, nausea, vomiting, vertigo, ataxia, drowsiness, and coma; instant collapse in infants after the local application of menthol/camphor to their nostrils. Seizures may be the first clinical sign of severe toxicity of camphor; however seizures are usually self-limited. Severe toxicity of camphor can result in delirium, visual hallucinations, cerebral edema, and status epilepticus. Systemic toxicity may include hypotension, tachycardia, respiratory failure and death.

***Treatment:***

The stomach should be emptied by gastric lavage. Salicylate remaining in the stomach may be absorbed by activated charcoal. Fluid and electrolyte management is the mainstay of treatment with the immediate aim of correction of acidosis, hyperpyrexia, hypokalaemia and dehydration. Alkaline diuresis, haemodialysis or haemoperfusion are effective methods of removing salicylate from the plasma.

Any convulsions must be controlled first through supportive care including anticonvulsant therapy.

**Packing**

Packed in plastic jars of 20 g, 30 g and 50 g.

**Storage Condition**

Store below 30°C in a dry place, protected from direct light.  
Keep away from reach of children

FOR EXTERNAL USE ONLY

**Expiry Date**

2 years from date of manufacture.

**Name and Address of Product Distributors:**

**ZONTRON PHARMACEUTICALS SDN BHD** (445695-T)

Lot 10 & 11, PERDA Industrial Park,  
Lorong IKS Simpang Ampat B,  
14100 Simpang Ampat, S.P.S.,  
Pulau Pinang, Malaysia

**Name and Address of Product Registration Holder /Manufacturer:**

**TERAPUTICS SDN. BHD.** (590500-W)

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**Malaysian Drug Registration No.** MAL19950124XC

**Date of Revision:** 10/06/2020